Ex:No: 8 Date: 15/4/26	Iris Flower Clustering Using K-Means Algorithm

Problem Description:

The Iris dataset contains 150 samples of three iris species (**Setosa, Versicolor, Virginica**) with four numerical features: **Sepal Length, Sepal Width, Petal Length, Petal Width**. The goal is to **group the flowers into clusters** based on these features using the **K-Means clustering algorithm**, without using the species labels.

Objective:

The objective of this program is to implement the K-Means clustering algorithm in R to group iris flowers based on their numerical attributes, visualize the clusters, and evaluate clustering performance against the actual species.

Procedure:

- **Step 1: Install and Load Required Libraries**

  We install and load the necessary libraries for implementing K-Means clustering.

```
# Install and Load required libraries
install.packages("factoextra")
library(ggplot2)
library(cluster)
library(factoextra)
```

- **Step 2: Load and Prepare the Dataset**

  We use the iris dataset (excluding the Species column) for clustering.

```
# Load the iris dataset
data(iris)
```

```
# Remove the Species column (only numeric features)
iris_data <- iris[,-5]
```

```
# Display first few rows
head(iris_data)
```

OUTPUT:

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
5	5.0	3.6	1.4	0.2
6	5.4	3.9	1.7	0.4

- **Step 3: Determine the Optimal Number of Clusters (k)**

  We use the Elbow Method to find the optimal number of clusters.

  # Elbow method to find optimal k using fviz_nbclust()

  ```
  fviz_nbclust(iris_data, Kmeans, method = "wss")+
  ggtitle("Elbow method for optimal k")
  ```

  Interpretation:

  The optimal k is chosen where the within-cluster sum of squares (WSS) starts to level off (elbow point).

- **Step 4: Apply K-Means Clustering Algorithm**

  We apply K-Means clustering with k = 3 (since iris dataset has 3 species).

  # Apply K-Means clustering with 3 clusters

  ```
  Kmeans_model <- Kmeans(iris_data, centers=3,
                         nstart=25)
  ```

  # Print cluster centers

  ```
  print(Kmeans_model$centers)
  ```

  # Print cluster assignments

  ```
  print(Kmeans_model$cluster)
  ```

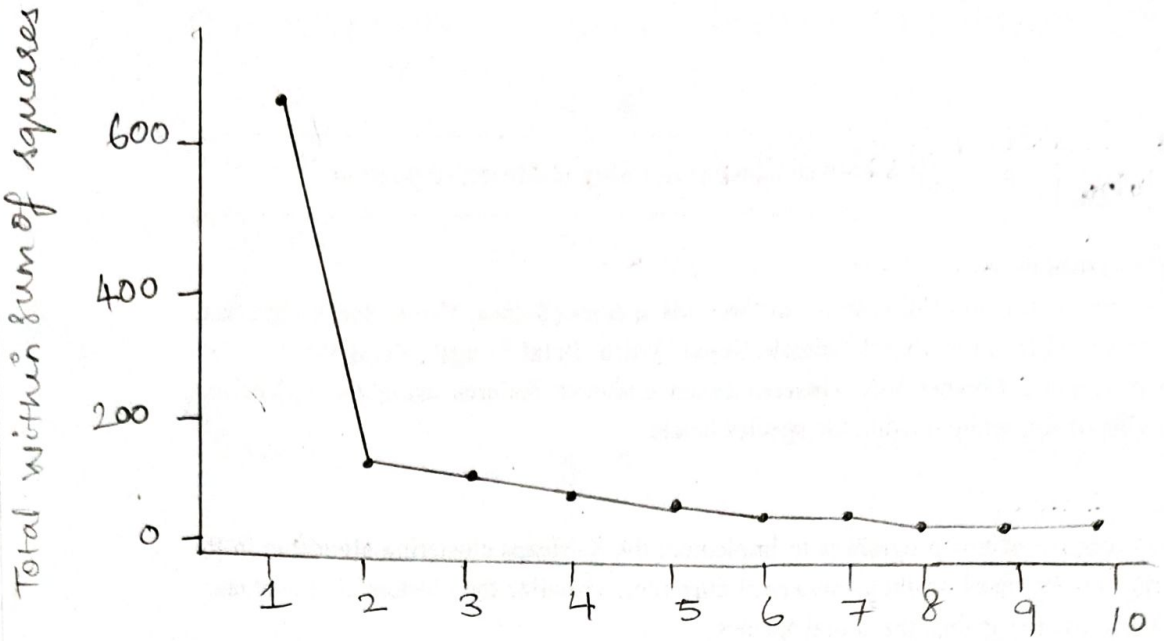
- **Step 5: Visualize Clusters**

  We plot the clusters using Principal Component Analysis (PCA) for dimensionality reduction.

  # Visualize clusters - fviz_cluster()

  ```
  fviz_cluster(Kmeans_model, data = iris_data
    geom = "point", ellipse.type = "convex") +
  ggtitle(" KMeans clustering on Iris Dataset")
  ```

Elbow method for optimal K



->	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	5.006000	3.428000	1.462000	0.246000
2	5.901613	2.748387	4.393548	1.433871
3	6.850000	3.073684	5.742105	2.071053

-> [1] 1 1 1

- **Step 6: Evaluate Cluster Performance**

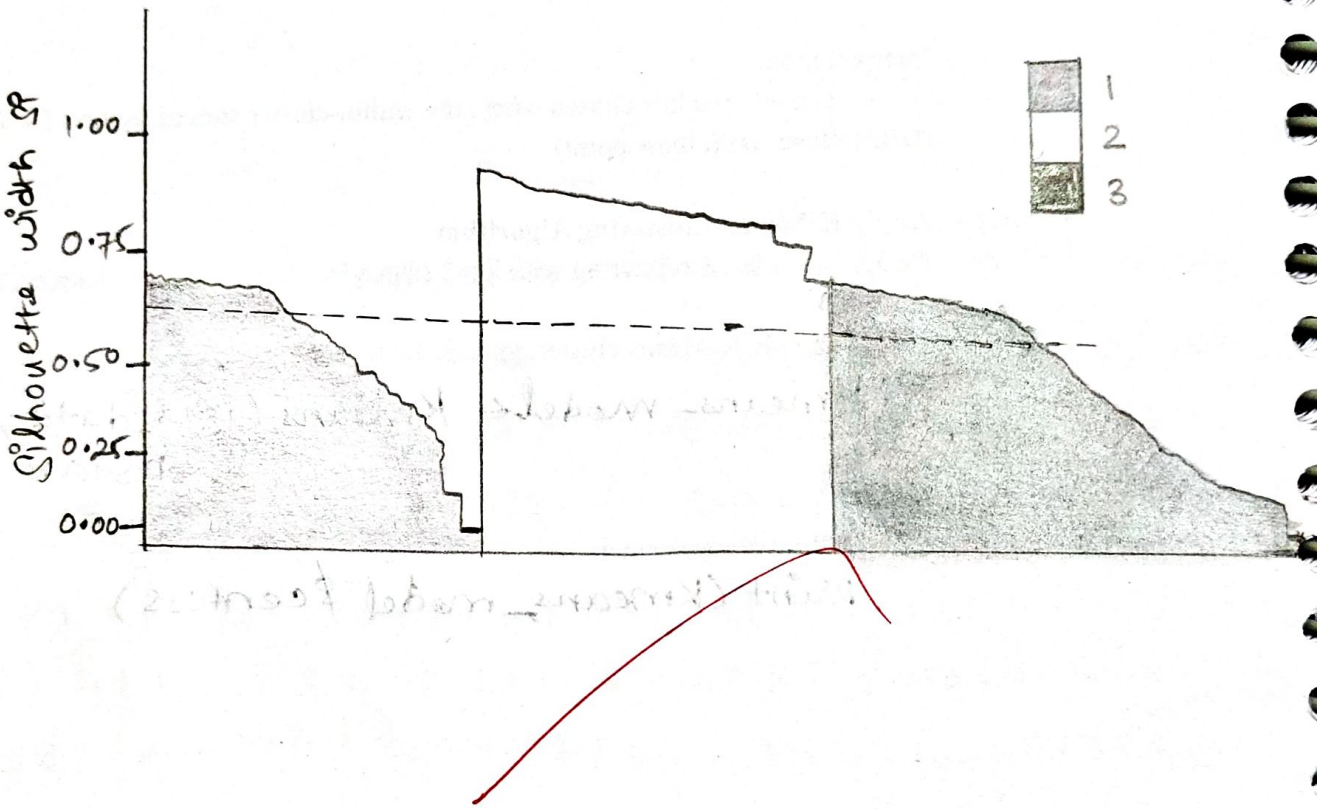
  We calculate the Silhouette Score to measure clustering quality.

```
# Compute silhouette score
silhouette_score <- silhouette(kmeans_model$cluster, dist(iris_data))
```

```
# Plot silhouette analysis
fviz_silhouette(silhouette_score)
```

Result:

Thus, the K-Means clustering algorithm was successfully implemented in R, and the resulting clusters were visualized and evaluated, effectively revealing the natural groupings in the Iris dataset.

	cluster	size	ave. sil. width
1	1	50	0.80
2	2	62	0.42
3	3	38	0.45

clusters silhouette plot
Average silhouette width : 0.55